Lecture 6: Addendum

Dude, Where's My Metadata?

Data must be "published"

Journals now (thankfully) demand that data is published in an "official" repository.

Yet the process is still fraught with many pitfalls.

Data is meaningless without knowing what it represents -> Metadata

What is in the sample named `XYJSLL8947878373`?

You'd would think that must *of course* be solved in a simple and logical way...

... you'd be wrong ...

Where is the data ...

You could publish it via:

- **SRA** - Short Read Archive https://www.ncbi.nlm.nih.gov/sra

or

- **GEO** - Gene Expression Omnibus https://www.ncbi.nlm.nih.gov/geo/

Both from NCBI ... the very same organization starts out by offering two different ways to store data.

Why?

Giants Battling out for Supremacy

You just fell into the battle of bureaucracy



Both **SRA** and **GEO** wants to be **TOO BIG TO FAIL**

Unlikely Truce ...



- **SRA** (Short Read Archive) will store the sequencing data
- **GEO** (Gene Expression Omnibus) will store what each sequencing data means

Both sites are antiquated and inadequate

What does that mean for you?

Extra work.

More chances to get it wrong.

Artifical barriers for science.

How does that look in practice?

Take this publication:

- A comparison of the sexually dimorphic dexamethasone transcriptome in mouse cerebral cortical and hypothalamic embryonic neural stem cells in *Molecular and Cellular Endocrinology (2018)*

Where is the data?

A publication may cite a SRA or GEO number

- **SRA** -> BioProject that starts with `PRNJ`
- **GEO** -> Gene Expression Series number `GSE`

This publication is

> *The raw and processed data was submitted to GEO (accession #GSE95363)*

Ok so they chose GEO.

Visite the GEO project

Note how you can click about

on the website. Follow seemingly circular links forever

Reproducibly connecting the files

Connecting **GEO Series** `GSE95363` to **SRA BioProject**

```
esearch -db gds -query GSE95363 | elink -target sra |
                                    efetch -format runinfo
```

will return a file where the *1st* and *22nd* columns are

```
Run,BioProject
SRR5285036,PRJNA376745
SRR5285037,PRJNA376745
SRR5285038,PRJNA376745
...
```

Ha! So `PRJNA376745` will contain data for `GSE95363`

Note how SRA has missing information

Essential information is missing at SRA

```
esearch -db gds -query GSE95363 | elink -target sra |
                        efetch -format runinfo > runinfo.txt
cat runinfo.txt
| csvcut -c Run,BioProject,Disease,Body_Site
```

Contains

```
Run,BioProject,Body_Site
SRR5285036,PRJNA376745,,
SRR5285037,PRJNA376745,,
...
```

The missing info is at GEO .. gee thanks NCBI!

Why not search directly at GEO

```
esearch -db gds -query GSE95363 | efetch
```

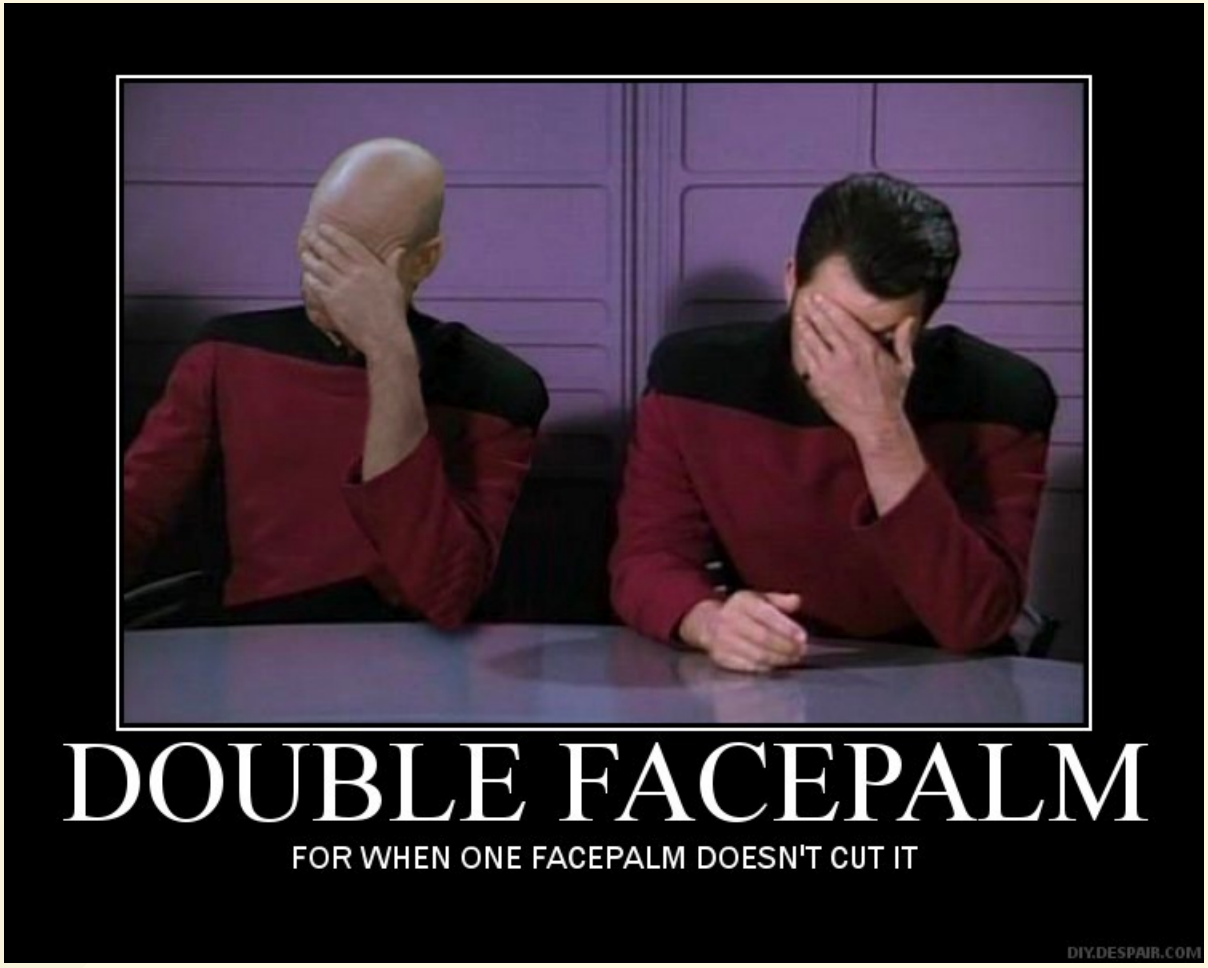
It will produce entries like:

```
3. Male ethanol vehicle treatment, hypothalamus, biologica
Organism:       Mus musculus
Source name:    Male ethanol vehicle treatment, Hypothalam
Platform: GPL17021 Series: GSE95363
FTP download:
SRA Run Selector: https://www.ncbi.nlm.nih.gov/Traces/stud
Sample          Accession: GSM2508044   ID: 302508044
```

This list does not contain the BioProject Run Number

...

Thank you NCBI for the "leadership"



How did we get here?

For a few samples, it is not a big deal to manually copy paste.

For dozens or hundreds samnples -- it is a massive and growing problem.

The current processes make it too easy to produce the most critical problem of all sciences --> **mislabeled data**.

Solutions?

A cottage industry of "simple" tools and techniques that should not even exist.

Slows down reproducibilty right from the get go.

- Write your own mini program.
- Search the web for a solution.
- Copy-paste manually into a new file.

All it would take a simple file to connect the two in a tabular way.

On the bright side.

This is why bioinformatics is much better paid job

Our leaders set this up such that that it take a lot of work just to figure out what the data is...